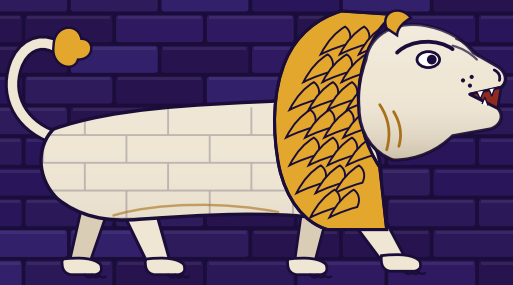
PHYSICS & ENERGY TECHNOLOGY · LEVEL 3

MH

MECHANICS AND HEAT

A handout of ideas, facts and problems



10 sections · Edition of November 2026 · 33 pages

Contents

How to read this handout . . . 1

Mechanics and Heat . . . 2

1 Choose units and describe motion . . . 2
2 Constant acceleration and graphs . . . 3
3 Resolve vectors before adding . . . 4
4 Projectiles: one time, two directions . . . 6
5 Newton laws and friction . . . 7
6 Circular motion and connected objects . . . 9
7 Work and kinetic energy . . . 10
8 Energy conservation and springs . . . 11
9 Temperature, calorimetry and phase changes . . . 13
10 Heat transfer and thermal expansion . . . 14

Hints . . . 16

Answers . . . 17

Mechanics and Heat, Test A . . . 19

Mechanics and Heat, Test B . . . 21

Mechanics and Heat, Test A: answers . . . 23

Mechanics and Heat, Test B: answers . . . 24

Facts at a glance . . . 25

Notation . . . 27

Sources . . . 29

Index . . . 30

Index of notation . . . 31

How to read this handout

This handout follows the method of Jaan Kalda's physics handouts. The text is short; the work is in the numbered items. Read each section once, then return to its items until you can use them without looking.

Idea 1. An *idea* is a way of thinking that unlocks a whole family of problems. Ideas are framed in violet.

Fact 1. A *fact* is a law or relationship you are expected to know, with the meaning of every symbol and the conditions under which it holds.

Method 1. A *method* is a fixed sequence of steps for a standard task.

Pr 1. Problems are numbered through the handout. A problem followed by a *Solution* is worked for you; cover the solution and try first. For the remaining problems, attempt the problem, then use the *Hints* section, and only then the *Answers* section, both at the back of the subject.

Physical quantities are set in italic, units in roman, with a thin space between number and unit. The index at the end lists every topic and symbol with the page on which it is introduced.

Mechanics and Heat

Motion, vectors, forces, work, energy, temperature and heat. New merged lectures are the main mechanics source; the older heat lecture supplies thermal topics.

1 Choose units and describe motion

Aims. Distinguish distance from signed displacement; Calculate average speed and average velocity from a route.
Before this section: Add and subtract lengths; divide by elapsed time; convert minutes to seconds.

Position tells you where an object is. Displacement is final position minus initial position; distance counts the entire path. Average velocity uses displacement, while average speed uses distance. A negative velocity gives direction, not a negative speed.

Idea 1. Imagine laying a measuring tape along the whole route, then stretching a separate arrow directly from start to finish. The tape represents distance; the arrow represents displacement. Turning back adds tape length while shortening the arrow, so speed and velocity can describe the same journey differently.

Fact 1.

$$\bar{v} = \Delta x/\Delta t$$
$$\text{average speed} = \text{distance}/\Delta t$$

where Δx is the final position minus initial position (m); Δt is the elapsed time (s); $\bar{v}$ is the average velocity (m/s).

Why it holds. Distance adds each part of the route because every part was travelled. Displacement depends only on endpoints, so reversing direction changes its sign contribution.

Method 1.

1. Choose east or right as positive and sketch the route.
2. Add every path length for distance; subtract start position from finish for displacement.
3. Divide distance and displacement separately by the same total time.
4. Attach a direction to velocity and a unit to every result.

Recognising it. Use this distinction whenever a route includes a turn, a return or a request for both speed and velocity.

Pr 1. A student walks 30 m east and 10 m west in 20 s.

Solution. Take east as positive: the two legs are +30 m and −10 m. Distance counts both legs: 30 + 10 = 40 m; displacement is 30 − 10 = +20 m. Over 20 s, average speed is 40/20

$= 2$ m/s and average velocity is $+20/20 = +1$ m/s, east.

Common error. Do not replace displacement with distance in a velocity question.

In the examination. Write the two numerators before dividing: distance for speed, displacement for velocity. This earns method marks even if arithmetic slips.

Pr 2. Can average speed be nonzero while average velocity is zero?

Pr 3. A cart travels 12 m right then 4 m left in 4 s. Find distance, displacement, average speed and average velocity.

Pr 4. Two walkers start and finish together in the same elapsed time but take routes of different lengths. Which averages must match?

(a) Their average velocities

(b) Their average speeds

(c) Both averages

2 Constant acceleration and graphs

Aims. Select a constant-acceleration equation from the known quantities; Interpret slope and area on a velocity-time graph.
Before this section: Signed velocity; rectangle and triangle areas; rearranging a simple equation.

Acceleration measures the change of velocity per second. On a position-time graph, slope is velocity. On a velocity-time graph, slope is acceleration and signed area is displacement. The constant-acceleration equations require acceleration to remain constant.

Idea 2. Acceleration tells how much the velocity changes during each second. A velocity-time graph is like a running record of how quickly position accumulates: each thin time strip contributes velocity times time. The slope describes velocity change, whereas the accumulated signed area describes displacement.

Fact 2.

$$v = u + at$$
$$s = ut + \tfrac{1}{2}at^2$$
$$v^2 = u^2 + 2as$$

where u is the initial velocity (m/s); v is the final velocity (m/s); a is the constant acceleration (m/s^2); s is the displacement (m); t is the elapsed time (s).
Conditions. The equations in this lesson assume constant acceleration throughout the stated interval.

Why it holds. Constant acceleration makes velocity change evenly, so the velocity-time graph is a straight line. With positive forward velocity and negative acceleration, the car initially slows because the change opposes its motion.

Method 2.

1. List the known quantities with signs and identify the unknown.
2. Check that the acceleration is constant before using the equations.
3. Choose an equation without an extra unknown, then substitute with units.
4. Use graph slope or area as an independent interpretation of the result.

Recognising it. Look for a constant acceleration, a uniform slowing statement or a straight velocity-time segment.

Pr 5. A car starts from rest and accelerates at 3 m/s^2 for 4 s.

Solution. Starts from rest means $u = 0$; the example gives $a = 3\ m/s^2$ and $t = 4$ s. The velocity gain is $at = 12$ m/s, so $v = 12$ m/s. The displacement is $\frac{1}{2} \times 3 \times 4^2 = 24$ m. On the velocity-time graph the triangle has base 4 s and height 12 m/s; its area is also 24 m.

Common error. A straight horizontal velocity-time graph means zero acceleration, not zero velocity.

In the examination. For a graph, state which axes you used before calculating slope or area; their units identify acceleration or displacement.

Pr 6. A car has velocity +8 m/s and acceleration $-2\ m/s^2$. Is it initially speeding up?

Pr 7. A car slows from 20 to 8 m/s in 4 s with constant acceleration. Find acceleration and displacement.

Pr 8. A car moves forward while its velocity-time line slopes downward but stays above zero. What happens?

(a) It is stationary because acceleration is negative

(b) It moves forward and slows down

(c) It moves backward and speeds up

3 Resolve vectors before adding

Aims. Add perpendicular vectors using signed components; Give a resultant direction in the correct quadrant.
Before this section: Right-triangle trigonometry; Pythagoras; compass directions.

A vector has size and direction. Two vectors must be added through their signed components, not by simply adding magnitudes. The resultant points from the initial tail to the final head in a head-to-tail drawing.

Idea 3. Components are the separate instructions for moving horizontally and vertically. A component pair describes one arrow even when one instruction is negative. Magnitude measures the arrow length, while signs tell which quadrant it occupies; a calculator angle cannot replace that directional information.

Fact 3.

$$A_x = A\cos\theta$$
$$A_y = A\sin\theta$$
$$R = \sqrt{R_x^2 + R_y^2}$$

where A_x is the horizontal vector component (same as vector); A_y is the vertical vector component (same as vector); R is the resultant magnitude (same as vector); θ is the angle from the stated axis (degrees or radians).

Why it holds. Perpendicular components form a right triangle, so their squares combine to give magnitude. Squaring removes signs, so the quadrant must be recovered from the original component signs.

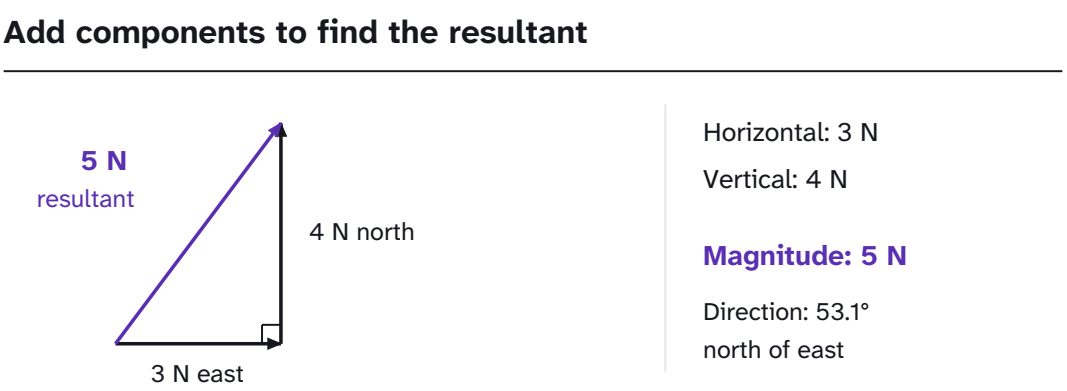


Figure 1. Figure 01. A 3 N east force and a 4 N north force give a 5 N resultant at 53.1° north of east. The force arrows use one common scale; the balancing force points in the opposite direction.

Method 3.

1. Draw axes and identify the reference axis for each angle.
2. Resolve each vector and assign signs to its components.
3. Add horizontal components and vertical components separately.
4. Calculate magnitude and use component signs to choose the direction.

Recognising it. Use components when vectors point in different directions, especially when an angle alone could be ambiguous.

Pr 9. A 3 N force east and a 4 N force north produce a 5 N resultant, 53.1 degrees north of east.

Solution. East and north are perpendicular, so the example has Rx = 3 N and Ry = 4 N. The magnitude is the square root of $3^2 + 4^2$, giving 5 N. The angle satisfies $\tan\theta = 4/3$, so $\theta = 53.1°$ north of east. A balancing force reverses both components: 3 N west and 4 N south, with the same 5 N magnitude.

Common error. The inverse tangent alone can return the wrong quadrant.

In the examination. Report both an angle and its reference direction; an angle alone can leave the quadrant ambiguous.

Pr 10. Does a vector with negative horizontal and positive vertical components point northeast?

Pr 11. Find the magnitude and direction of a vector with components -3 m and $+4$ m.

Pr 12. Two vectors have components $(3,4)$ and $(-3,4)$. What do they share?

(a) Equal direction but different magnitude

(b) Both magnitude and direction

(c) Equal magnitude but different direction

4 Projectiles: one time, two directions

Aims. Separate a projectile into horizontal and vertical motions; Use the shared flight time to calculate horizontal range.

Before this section: Constant-acceleration motion; squaring and square roots; horizontal and vertical directions.

Ignoring air resistance, horizontal acceleration is zero and vertical acceleration is downward. Horizontal and vertical motion share the same time. At the top, only the vertical velocity becomes zero; horizontal velocity continues.

Idea 4. Imagine filming a dropped ball and a horizontally launched ball from the side. With equal starting height and no air drag, their vertical motions match. The launched ball also moves sideways, but that sideways motion does not delay gravity; both directions share one clock.

Fact 4.

$$v_x = u\cos\theta$$
$$v_y = u\sin\theta - gt$$
$$y = u\sin\theta\, t - \tfrac{1}{2}gt^2$$

where u_x is the initial horizontal velocity (m/s); u_y is the initial vertical velocity (m/s); g is the gravitational acceleration magnitude (m/s^2); t is the flight time (s).
Conditions. Ignore air resistance and take gravity as uniform; launch and landing heights need not match.

Why it holds. Gravity changes vertical velocity, while zero horizontal force leaves horizontal velocity constant. Landing time is determined by the vertical fall; that same time determines the sideways range.

Method 4.

1. Draw the launch point and select a positive vertical direction.
2. Write horizontal motion with zero acceleration and vertical motion with gravity.
3. Use the height information to find the positive physical flight time.
4. Use this same time in the horizontal displacement equation.

Recognising it. A horizontal launch from a table or cliff signals a vertical free-fall calculation followed by horizontal motion.

Pr 13. A ball leaves a horizontal table at 3 m/s and takes 0.4 s to land.

Solution. A horizontal launch has initial vertical velocity zero; its horizontal velocity remains 3 m/s. The stated time is 0.4 s, so horizontal range is $3 \times 0.4 = 1.2$ m. The vertical drop is $\frac{1}{2} \times 9.8 \times 0.4^2 = 0.784$ m. The unequal range and drop are consistent because the two directions obey different motion equations.

Common error. Do not set the total velocity to zero at the highest point.

In the examination. If you choose upward positive, the downward displacement and acceleration must both carry negative signs.

Pr 14. At the highest point of an angled projectile, which velocity component is zero?

Pr 15. A ball is projected horizontally at 5 m/s from a height of 4.9 m. Ignore air resistance and use $g = 9.8$ m/s^2. Find flight time and horizontal range.

Pr 16. Two balls leave the same height at the same time, one dropped and one launched horizontally. Ignoring air drag, which lands first?

(a) They land together

(b) The horizontally launched ball lands later

(c) The dropped ball lands later

5 Newton laws and friction

Aims. Construct a free-body diagram for one selected object; Use net force and friction to find acceleration.
Before this section: Signed addition; mass and weight; perpendicular directions.

A free-body diagram shows only forces acting on the selected object. Net force causes acceleration. Action-reaction forces act on different objects and therefore do not cancel on one free-body diagram. Static friction adjusts up to its maximum; kinetic friction applies during sliding.

Idea 5. A free-body diagram is an inventory of pushes and pulls on one object. The object accelerates according to the unbalanced part of that inventory. Friction opposes the sliding direction, and the normal force adjusts to the perpendicular force balance; neither is an additional forward driving force.

Fact 5.

$$\sum F = ma$$
$$W = mg$$
$$f_k = \mu_k N$$
$$f_s \leq \mu_s N$$

where m is the object mass (kg); N is the normal contact force (N); μ_k is the coefficient of kinetic friction (dimensionless); a is the acceleration ($\mathrm{m/s^2}$).
Conditions. The example treats the block as sliding and the given 4 N as kinetic friction.

Why it holds. The horizontal pull and kinetic friction oppose, so only their difference produces horizontal acceleration. With a horizontal pull and no vertical acceleration, the floor normal balances weight in this particular setup.

Method 5.

1. Isolate one object and draw only the forces acting on it.
2. Choose axes and use the perpendicular force balance to obtain the normal force if needed.
3. Choose static or kinetic friction from the stated motion.
4. Add signed forces along the motion and divide the net force by mass.

Recognising it. Use a force diagram before calculating whenever a problem includes a pull, friction, contact or an incline.

Pr 17. A 2 kg block on a horizontal surface is pulled with 10 N.

Solution. Choose the 10 N pull as positive; the 4 N sliding friction acts in the opposite direction. The horizontal net force is $10 - 4 = 6$ N. For the 2 kg block, $a = 6/2 = 3\ \mathrm{m/s^2}$ in the direction of the pull. Weight and normal force balance vertically under the horizontal-surface assumptions, so they do not enter the horizontal sum.

Common error. The normal force is not always mg: angled pulls and inclines change it.

In the examination. Show the signed force equation before substitution; do not write the applied force alone as ma.

Pr 18. A sliding block has a 7 N pull and 7 N friction. Must it be at rest?

Pr 19. A 5 kg block slides to the right on a horizontal floor with coefficient of kinetic friction 0.2. It is pulled horizontally to the right with 20 N. Use $g = 10\ \mathrm{m/s^2}$. Find normal force, friction and acceleration.

Pr 20. A block already slides right. Its pull becomes smaller than the leftward kinetic friction. What happens initially?

(a) It accelerates right because it is moving right

(b) It keeps moving right but slows

(c) It immediately moves left

6 Circular motion and connected objects

Aims. Identify the real forces that supply inward acceleration; Set up separate force equations for connected masses.
Before this section: Newton second law; force diagrams; uniform circular motion directions.

Changing direction is acceleration even at constant speed. Centripetal means inward; it names the required net force, not an extra physical force. In an ideal taut string over a massless frictionless pulley, connected objects share acceleration magnitude.

Idea 6. Changing direction is a change in velocity even when speed stays fixed. For circular motion, the real forces must supply that inward change. For connected masses, internal string forces can be eliminated by considering the whole system, while separate diagrams are needed to recover the tension.

Fact 6.

$$a_r = v^2/r$$
$$\sum F_{\text{inward}} = mv^2/r$$

where v is the speed along the circle (m/s); r is the circle radius (m); a_r is the inward acceleration (m/s^2); T is the string tension (N).
Conditions. Connected-object formulas assume a taut massless string and an ideal frictionless pulley when stated.

Why it holds. Tension acts on both connected masses, but it cancels from the whole-system driving-force calculation. The heavier side supplies the weight imbalance; both masses contribute inertia and therefore belong in the total mass.

Method 6.

1. For circular motion mark the centre and select inward as a positive radial direction.
2. Sum the actual radial force components and equate them to mv^2/r.
3. For connected objects draw a diagram for each mass and use compatible acceleration directions.
4. Use the ideal string constraint to share acceleration magnitude, then solve for acceleration and tension.

Recognising it. For circular paths look for the inward net force; for a pulley pair use one system equation and one mass equation.

Pr 21. A 2 kg object moves at 3 m/s in a circle of radius 1.5 m.

Solution. In the circular example, v = 3 m/s and r = 1.5 m. Squaring speed first gives ar = 9/1.5 = 6 m/s^2, directed toward the centre. Multiplying by the 2 kg mass gives 12 N inward net force. The problem would need a physical setup to decide whether tension, gravity or another force provides those 12 N.

Common error. Do not add a separate centripetal force to tension or gravity.

In the examination. Do not put tension and an extra centripetal force on the same diagram; centripetal describes the inward net force.

Pr 22. If the speed remains fixed and radius doubles, what happens to inward acceleration?

Pr 23. An ideal Atwood machine has masses 2 kg and 3 kg. Use $g = 10\ \mathrm{m/s^2}$. Find acceleration and tension.

Pr 24. For an ideal Atwood machine, why is the total mass used in finding acceleration?

(a) Only the heavier mass moves

(b) Tension adds extra mass to the system

(c) Both connected masses must accelerate

7 Work and kinetic energy

Aims. Determine the sign of work from force and displacement directions; Calculate net work from a change in kinetic energy.
Before this section: Cosine of common angles; squaring speed; positive and negative energy transfers.

Work transfers energy when a force has a component along displacement. A perpendicular force does no work. Net work equals the change in kinetic energy. For a varying force along a line, work is the signed area under the force-position graph.

Idea 7. Work measures how a force transfers energy along a displacement. Carrying a force at an angle is like splitting it into a useful along-path component and a sideways component. Net work concerns all forces together, so it is directly linked to the change in speed through kinetic energy.

Fact 7.

$$W = Fd\cos\theta$$
$$K = \tfrac{1}{2}mv^2$$
$$W_{\text{net}} = K_f - K_i$$

where W is the work transferred by a force (J); F is the force magnitude (N); d is the displacement magnitude (m); K is the kinetic energy (J).

Why it holds. Only the component parallel to displacement contributes to work because the perpendicular component has no along-path displacement. Kinetic energy depends on speed squared, so equal speed increases at different starting speeds require different net work.

Method 7.

1. Identify the displacement and the angle between it and the force.
2. Use only the force component along displacement to calculate work.
3. Add work from all forces for net work, or use the kinetic-energy change.
4. Check that positive net work corresponds to an increase in speed.

Recognising it. Use the work-energy theorem when speeds and mass are given but the detailed force or time history is unnecessary.

Pr 25. A 10 N force pulls through 3 m at 60 degrees to displacement.

Solution. The example states an angle of 60° between the 10 N force and 3 m displacement. Only $10 \cos 60° = 5$ N acts along the displacement. Work by this force is $5 \times 3 = 15$ J. The remaining perpendicular component contributes zero work; 15 J is not automatically net work if other forces also do work.

Common error. Use the force-displacement angle, not an unrelated angle in the drawing.

In the examination. When using the energy theorem, square each speed before subtracting; $(\mathrm{vf} - \mathrm{vi})^2$ is not $\mathrm{vf}^2 - \mathrm{vi}^2$.

Pr 26. Can a nonzero force do zero work during a displacement?

Pr 27. A 2 kg object speeds up from 3 to 5 m/s. Find the net work done.

Pr 28. The same object goes from speed 1 to 2 m/s, then from 2 to 3 m/s. Which stage requires more net work?

(a) The second stage

(b) The first stage

(c) Both require equal net work

8 Energy conservation and springs

Aims. Track kinetic, gravitational and spring energy between two states; Identify when a loss term is required in an energy balance.
Before this section: Work and kinetic energy; choosing a height reference; spring compression or extension.

Energy can move between kinetic, gravitational and elastic stores. Mechanical energy stays constant only when the model excludes losses or other transfers. Choose a height reference; only height changes affect the result.

Idea 8. Energy bookkeeping compares two snapshots rather than following every instant of motion. A compressed spring initially stores energy; when it returns to its natural length, that store can become kinetic energy. The exchange is exact only after accounting for any friction, air drag or other transfer.

Fact 8.

$$K_i + U_i + W_{\text{other}} = K_f + U_f$$
$$U_g = mgh$$
$$U_s = \tfrac{1}{2}kx^2$$

where U_g is the gravitational potential energy (J); U_s is the elastic potential energy (J); k is the spring stiffness (N/m); x is the spring deformation from natural length (m).
Conditions. The falling-mass example neglects air resistance and other energy transfers; spring-energy formulas require the proportional linear range.

Why it holds. At natural length the ideal spring deformation is zero, so its elastic energy term vanishes. On a level frictionless surface gravitational energy does not change, allowing spring energy to become block kinetic energy.

Method 8.

1. Name initial and final states and choose a zero-height level.
2. Write the energy balance before removing terms that are zero.
3. Include friction or other transfers if the model contains them.
4. Solve for the unknown and check the physical sign and units.

Recognising it. Choose an energy balance when initial and final positions or spring deformations are given and time is not needed.

Pr 29. A mass drops 5 m from rest without losses.

Solution. Set the final height to zero; the initial height is then 5 m. The object starts from rest, so its initial energy is gravitational: mg × 5. With no losses, mg × 5 equals one half mv^2; dividing by mass shows why mass cancels. Using g = 10 m/s^2 gives $v^2 = 100$ and speed v = 10 m/s.

Common error. Spring extension is measured from natural length, not total length.

In the examination. For spring energy convert centimetres to metres before squaring the deformation.

Pr 30. Would the same loss-free 5 m fall give a heavier object a larger final speed?

Pr 31. A spring of constant 200 N/m is compressed by 0.10 m and releases a 0.50 kg block on a frictionless horizontal surface. Find its speed when the spring returns to natural length.

Pr 32. The same compressed spring releases two blocks separately on a frictionless surface. Which block leaves faster?

(a) Both have the same speed

(b) The lighter block

(c) The heavier block

9 Temperature, calorimetry and phase changes

Aims. Separate temperature change from phase change in a heating problem; Apply an energy balance with consistent mass and temperature units.

Before this section: Energy in joules; mass conversion; subtraction of temperatures.

Temperature describes thermal state; heat is energy transferred because of a temperature difference. During a phase change of a pure substance at fixed pressure, energy can transfer without a temperature change. In an insulated mixing model, heat lost equals heat gained.

Idea 9. Specific heat tells how much energy a material needs for a temperature rise. Latent heat describes a different job: changing phase. A heating problem becomes manageable when treated as a sequence of stages, each assigned the relation that matches what is physically changing.

Fact 9.

$$Q = mc\Delta T$$
$$Q = mL$$
$$T_K = T_C + 273.15$$

where Q is the energy transferred as heat (J); c is the specific heat capacity (J/(kg K)); L is the specific latent heat (J/kg); ΔT is the temperature change (K or °C difference).

Why it holds. A liquid that warms without a phase change needs energy proportional to its mass and temperature rise. The Celsius and kelvin scales have equally sized intervals, so a temperature difference has the same numerical value in both.

Method 9.

1. Identify the initial and final phase and temperature.
2. Split the process into warming, phase-change and further-warming stages where needed.
3. Use mcΔT for each warming stage and mL for each phase change.
4. Add the energy requirements or balance heat gained and lost in an insulated mixture.

Recognising it. Use specific heat when temperature changes within one phase; add latent heat only when the process crosses a phase transition.

Pr 33. Heating 0.20 kg of water by 10 degrees C needs Q = 0.20 × 4200 × 10 = 8400 J.

Solution. The water remains liquid while warming, so the relevant relation is Q = mcΔT. A rise of 10°C is a temperature difference of 10 K. For 0.20 kg and c = 4200 J/(kg K), Q = 0.20 × 4200 × 10 = 8400 J. If the process also melts ice, that extra stage needs mL; it is not included in the 8400 J.

Common error. A temperature difference of 10 degrees C equals 10 K; an absolute temperature does not.

In the examination. On a heating curve label sloping segments mcΔT and horizontal phase-change segments mL.

Pr 34. Does liquid water warming from 20°C to 30°C have ΔT = 283.15 K?

Pr 35. How much energy heats 0.50 kg of water from 20 to 30 degrees C? Use c = 4200 J/(kg K).

Pr 36. Two samples of the same liquid have the same temperature rise. One has twice the mass. How do their required heats compare?

(a) Both need the same energy

(b) The larger sample needs four times the energy

(c) The larger sample needs twice the energy

10 Heat transfer and thermal expansion

Aims. Classify heat transfer as conduction, convection or radiation; Predict rate changes and calculate small linear expansion.

Before this section: Direct and inverse proportionality; temperature differences; metres and millimetres.

Conduction transfers energy through matter without bulk transport; convection uses fluid motion; radiation uses electromagnetic waves and can cross vacuum. Most solids expand on warming. A thermal insulator reduces transfer but does not create cold.

Idea 10. Thermal expansion is a small fractional change spread over an original length. A longer rod contains more length over which the same fractional change acts. Heat-transfer equations describe energy flow instead; first decide whether the question asks for a size change or a rate of energy transfer.

Fact 10.

$$\Delta L = \alpha L_0 \Delta T$$
$$P_{\text{cond}} = kA\Delta T/L$$
$$P_{\text{rad}} = \epsilon\sigma A(T^4 - T_s^4)$$

where P is the heat-transfer rate (W); k is the thermal conductivity (W/(m K)); A is the transfer surface area (m^2); L is the wall thickness (m); α is the linear expansion coefficient (K^{-1}).

Conditions. The wall comparison uses steady one-dimensional conduction with unchanged material and temperature difference.

Why it holds. For fixed material, expansion grows with both original length and temperature rise. The coefficient gives fractional expansion per degree, so its unit is inverse

kelvin rather than metres.

Method 10.

1. Identify the physical mechanism and select the matching relation.
2. For a comparison, hold every unstated quantity fixed and change only the named factor.
3. For expansion use the original length and the temperature rise.
4. Use kelvin for absolute temperatures in radiation and convert the final length if requested.

Recognising it. Words such as rod length, expansion coefficient and temperature rise indicate a linear-expansion calculation.

Pr 37. Doubling the thickness of an otherwise unchanged wall halves its conductive heat-transfer rate.

Solution. For conduction, the same wall material and temperature difference give P proportional to A/L. Doubling thickness replaces L by 2L, so P becomes P/2. Doubling area instead gives 2P. If both area and thickness double together, the factors cancel and the rate is unchanged.

Common error. Use kelvin in fourth-power radiation equations; Celsius produces incorrect results.

In the examination. For a ratio question explicitly state what stays constant; otherwise the proportionality is incomplete.

Pr 38. A wall is made three times thicker with all other quantities unchanged. What is its new conduction rate?

Pr 39. A rod is 2 m long and has expansion coefficient 0.000012 per K. It warms by 50 K. Find the increase in length in metres and millimetres.

Pr 40. Two equal-length rods undergo the same temperature rise. The first has twice the expansion coefficient. What follows?

(a) The first length increase is twice as large

(b) Their length increases must match

(c) The first final length is twice as large

Hints

Use one hint at a time, then return to the problem.

Pr 3 (p. 3) (1) Sketch the two legs and choose right as positive. Which quantity adds lengths and which adds signed changes? (2) Use 12 m and 4 m positively for distance; use a negative contribution for the leftward leg in displacement. (3) Divide each separate total by 4 s. Keep the direction on velocity and check that speed is nonnegative.

Pr 7 (p. 4) (1) List initial velocity, final velocity and elapsed time. Decide the sign of the velocity change before calculating. (2) Find acceleration from final minus initial velocity divided by 4 s. (3) For displacement use the average of 20 and 8 m/s multiplied by 4 s; this is the area of the trapezoid under the graph.

Pr 11 (p. 6) (1) Place the horizontal component to the left and the vertical component upward. Identify the quadrant first. (2) Use the right triangle for length and find the acute reference angle from the ratio 4/3. (3) Express that angle north of west, or convert it to a counterclockwise angle from positive x. State both magnitude and direction.

Pr 15 (p. 7) (1) Separate the height information from the sideways speed. A horizontal launch has zero initial vertical velocity. (2) Use the fall-distance relation with height 4.9 m and gravitational acceleration 9.8 $\mathrm{m/s^2}$ to find the positive time. (3) Multiply that flight time by 5 m/s for range. Do not put the horizontal speed into the fall equation.

Pr 19 (p. 8) (1) Draw weight, normal force, the rightward pull and leftward sliding friction. Separate vertical from horizontal equations. (2) Use vertical balance to obtain normal force, then multiply it by the kinetic coefficient 0.2. (3) Subtract friction from the 20 N pull and divide the net force by 5 kg. State the acceleration direction.

Pr 23 (p. 10) (1) Decide which mass descends. Draw one force diagram for each mass without adding any extra driving force. (2) Use the difference of the two weights as the system driving force and the sum of masses as system inertia. (3) After finding acceleration, apply the upward force equation to the rising 2 kg mass to obtain tension; it is not simply either weight.

Pr 27 (p. 11) (1) This question gives mass and two speeds rather than a force and distance. Which energy theorem connects them directly? (2) Calculate final and initial kinetic energies separately, retaining each squared speed. (3) Subtract the initial energy from the final energy. A speed increase should give positive net work in joules.

Pr 31 (p. 12) (1) Identify the initial energy store and the final energy store. Is there friction or a change in height? (2) Set the initial spring energy equal to the final block kinetic energy, using the compression in metres. (3) Insert spring constant 200 N/m, compression 0.10 m and mass 0.50 kg. Solve for the positive speed after isolating its square.

Pr 35 (p. 14) (1) Check whether the water changes phase. Identify the initial and final temperatures rather than using either as the temperature rise. (2) Subtract 20°C from 30°C and use the given liquid-water specific heat. (3) Multiply mass 0.50 kg, specific heat 4200 J/(kg K) and the temperature difference. Report energy in joules.

Pr 39 (p. 15) (1) Identify original length and temperature rise; the question asks for the increase rather than the final length. (2) Multiply coefficient by original length 2 m and temperature rise 50 K. (3) The product is in metres. Multiply by 1000 to express the same increase in millimetres; do not add the original length unless asked.

Answers

Pr 2 (p. 3) Yes. A return trip has zero displacement but a positive total distance.

Pr 3 (p. 3) Distance = 16 m; displacement = +8 m; speed = 16/4 = 4 m/s; velocity = 8/4 = +2 m/s. Award one mark for each quantity.

Pr 4 (p. 3) (a) Correct. Matching endpoints give equal displacement and matching times give equal average velocity.

Pr 6 (p. 4) No. Acceleration opposes its current velocity, so its speed initially decreases.

Pr 7 (p. 4) a = (8−20)/4 = −3 m/s^2. s = [(20+8)/2] × 4 = 56 m. Two marks for each calculation, including the sign and unit.

Pr 8 (p. 4) (b) Correct. Positive velocity means forward motion; a decreasing positive velocity means lower speed.

Pr 10 (p. 6) No. It points northwest, in quadrant II.

Pr 11 (p. 6) Magnitude = 5 m. The vector is in quadrant II: 126.9 degrees counterclockwise from positive x, or 53.1 degrees north of west. Two marks for magnitude and two for a correct direction.

Pr 12 (p. 6) (c) Correct. Squared components give equal lengths, while horizontal signs place them in different quadrants.

Pr 14 (p. 7) Only the vertical component; horizontal velocity remains constant in this model.

Pr 15 (p. 7) 4.9 = 0.5 × 9.8 times t^2, so t = 1 s. Range = 5 × 1 = 5 m. Two marks per result with the equation shown.

Pr 16 (p. 7) (a) Correct. They share initial vertical velocity and vertical acceleration, so their fall times match.

Pr 18 (p. 8) No. Zero net force gives zero acceleration; it may continue moving at constant velocity.

Pr 19 (p. 8) N = 50 N, friction = 0.2 × 50 = 10 N, net force = 10 N, a = 10/5 = 2 m/s^2. One mark for each step.

Pr 20 (p. 8) (b) Correct. Velocity initially remains rightward, but the leftward net force reduces that velocity.

Pr 22 (p. 10) It halves because v^2/r is inversely proportional to radius at fixed speed.

Pr 23 (p. 10) Net driving force on the two-mass system is (3−2) × 10 = 10 N. Total mass = 5 kg, so a = 2 m/s^2. For the rising 2 kg mass, T−20 = 2 × 2, hence T = 24 N. Two marks per result.

Pr 24 (p. 10) (c) Correct. Both masses share an acceleration magnitude, so both contribute to the system inertia.

Pr 26 (p. 11) Yes. A force perpendicular to the displacement does zero work.

Pr 27 (p. 11) Net work = 0.5 × 2 × (25−9) = 16 J. Give marks for the energy theorem, both squared speeds, substitution and unit.

Pr 28 (p. 11) (a) Correct. The squared-speed increase is larger during the second stage.

Pr 30 (p. 12) No. Mass cancels from mgh = one half mv^2.

Pr 31 (p. 12) Spring energy = 0.5 × 200 × 0.10 squared = 1 J. Set 1 = 0.5 × 0.50 × v squared to get v = 2 m/s. Two marks for each stage.

Pr 32 (p. 12) (b) Correct. The same spring energy gives a larger speed when shared by a smaller mass.

Pr 34 (p. 14) No. $\Delta T = 10$ K. The Celsius-to-kelvin offset cancels when subtracting two temperatures.

Pr 35 (p. 14) Temperature rise = 10 K. Q = mc times temperature rise = 0.50 × 4200 × 10 = 21000 J. Four marks: model, temperature difference, substitution, result/unit.

Pr 36 (p. 14) (c) Correct. With unchanged specific heat and temperature rise, heat is proportional to mass.

Pr 38 (p. 15) One third of the original rate.

Pr 39 (p. 15) Increase = 0.000012 × 2 × 50 = 0.0012 m = 1.2 mm. Award marks for formula, substitution, metres and conversion.

Pr 40 (p. 15) (a) Correct. Length increase is proportional to the coefficient with the other factors fixed.

Mechanics and Heat, Test A

25 marks. Suggested time 25 minutes. Multiple-choice questions carry 1 mark each, written problems 4 marks each. Work without notes.

Part I. Choose one answer.

1. A runner completes one lap and returns to the start. Average velocity is:

   (a) Always positive
   (b) Impossible to define
   (c) Zero
   (d) Equal to average speed

2. The area under a velocity-time graph gives:

   (a) Displacement
   (b) Force
   (c) Speed at the final instant
   (d) Acceleration

3. Two perpendicular forces of 6 N and 8 N have resultant magnitude:

   (a) 10 N
   (b) 48 N
   (c) 2 N
   (d) 14 N

4. At the top of a projectile path, acceleration is:

   (a) Zero
   (b) Horizontal
   (c) Upward
   (d) Downward with magnitude g

5. An object moving at constant velocity has:

   (a) A forward net force
   (b) Increasing acceleration
   (c) Zero net force
   (d) No forces acting at all

Part II. Show your reasoning.

1. A cart travels 12 m right then 4 m left in 4 s. Find distance, displacement, average speed and average velocity.

2. A car slows from 20 to 8 m/s in 4 s with constant acceleration. Find acceleration and displacement.

3. Find the magnitude and direction of a vector with components −3 m and +4 m.

4. A ball is projected horizontally at 5 m/s from a height of 4.9 m. Ignore air resistance and use $g = 9.8\ \text{m/s}^2$. Find flight time and horizontal range.

5. A 5 kg block slides to the right on a horizontal floor with coefficient of kinetic friction 0.2. It is pulled horizontally to the right with 20 N. Use $g = 10\ \text{m/s}^2$. Find normal force, friction and acceleration.

Mechanics and Heat, Test B

25 marks. Suggested time 25 minutes. Multiple-choice questions carry 1 mark each, written problems 4 marks each. Work without notes.

Part I. Choose one answer.

1. Doubling speed at fixed radius changes centripetal acceleration by a factor of:

   (a) Two
   (b) Four
   (c) Eight
   (d) One half

2. A force perpendicular to displacement does work equal to:

   (a) Fd
   (b) Zero
   (c) -Fd
   (d) Half of Fd

3. If the extension of a Hooke-law spring doubles while k stays constant, its stored energy becomes:

   (a) Four times as large
   (b) Twice as large
   (c) Unchanged
   (d) Half as large

4. During melting at fixed pressure, added heat mainly:

   (a) Raises temperature continuously
   (b) Changes phase
   (c) Decreases mass
   (d) Disappears

5. Which heat-transfer mechanism can operate through a vacuum?

   (a) Fluid mixing
   (b) Convection
   (c) Conduction through air
   (d) Radiation

Part II. Show your reasoning.

1. An ideal Atwood machine has masses 2 kg and 3 kg. Use $g = 10\ \mathrm{m/s^2}$. Find acceleration and tension.

2. A 2 kg object speeds up from 3 to 5 m/s. Find the net work done.

3. A spring of constant 200 N/m is compressed by 0.10 m and releases a 0.50 kg block on a frictionless horizontal surface. Find its speed when the spring returns to natural length.

4. How much energy heats 0.50 kg of water from 20 to 30 degrees C? Use c = 4200 J/(kg K).

5. A rod is 2 m long and has expansion coefficient 0.000012 per K. It warms by 50 K. Find the increase in length in metres and millimetres.

Mechanics and Heat, Test A: answers

Suggested marking points are for self-study; your lecturer may use a different scheme.

Question 1. (c) The net displacement is zero even though the distance travelled is not.

Written part. Distance = 16 m; displacement = +8 m; speed = 16/4 = 4 m/s; velocity = 8/4 = +2 m/s. Award one mark for each quantity. (Section: p. 2.)

Question 2. (a) Area has units (m/s) × s = m and includes the sign of velocity.

Written part. a = (8−20)/4 = −3 m/s^2. s = [(20+8)/2] × 4 = 56 m. Two marks for each calculation, including the sign and unit. (Section: p. 3.)

Question 3. (a) The Pythagorean calculation gives the square root of 36+64 = 10 N.

Written part. Magnitude = 5 m. The vector is in quadrant II: 126.9 degrees counterclockwise from positive x, or 53.1 degrees north of west. Two marks for magnitude and two for a correct direction. (Section: p. 4.)

Question 4. (d) Gravity continues to act at the top.

Written part. 4.9 = 0.5 × 9.8 times t^2, so t = 1 s. Range = 5 × 1 = 5 m. Two marks per result with the equation shown. (Section: p. 6.)

Question 5. (c) Constant velocity means zero acceleration, so forces balance.

Written part. N = 50 N, friction = 0.2 × 50 = 10 N, net force = 10 N, a = 10/5 = 2 m/s^2. One mark for each step. (Section: p. 7.)

Mechanics and Heat, Test B: answers

Suggested marking points are for self-study; your lecturer may use a different scheme.

Question 1. (b) The speed is squared.

Written part. Net driving force on the two-mass system is $(3-2) \times 10 = 10$ N. Total mass = 5 kg, so a = 2 m/s^2. For the rising 2 kg mass, T$-$20 = 2 × 2, hence T = 24 N. Two marks per result. (Section: p. 9.)

Question 2. (b) The cosine of 90 degrees is zero.

Written part. Net work = 0.5 × 2 × (25$-$9) = 16 J. Give marks for the energy theorem, both squared speeds, substitution and unit. (Section: p. 10.)

Question 3. (a) Spring energy is proportional to extension squared.

Written part. Spring energy = 0.5 × 200 × 0.10 squared = 1 J. Set 1 = 0.5 × 0.50 × v squared to get v = 2 m/s. Two marks for each stage. (Section: p. 11.)

Question 4. (b) Energy changes the phase while temperature remains at the melting point.

Written part. Temperature rise = 10 K. Q = mc times temperature rise = 0.50 × 4200 × 10 = 21000 J. Four marks: model, temperature difference, substitution, result/unit. (Section: p. 13.)

Question 5. (d) Electromagnetic radiation needs no material medium.

Written part. Increase = 0.000012 × 2 × 50 = 0.0012 m = 1.2 mm. Award marks for formula, substitution, metres and conversion. (Section: p. 14.)

Facts at a glance

1. Choose units and describe motion p. 2

$$\bar{v} = \Delta x / \Delta t$$
$$\text{average speed} = \text{distance}/\Delta t$$

2. Constant acceleration and graphs p. 3

$$v = u + at$$
$$s = ut + \tfrac{1}{2}at^2$$
$$v^2 = u^2 + 2as$$

3. Resolve vectors before adding p. 4

$$A_x = A\cos\theta$$
$$A_y = A\sin\theta$$
$$R = \sqrt{R_x^2 + R_y^2}$$

4. Projectiles: one time, two directions p. 6

$$v_x = u\cos\theta$$
$$v_y = u\sin\theta - gt$$
$$y = u\sin\theta\, t - \tfrac{1}{2}gt^2$$

5. Newton laws and friction p. 7

$$\sum F = ma$$
$$W = mg$$
$$f_k = \mu_k N$$
$$f_s \le \mu_s N$$

6. Circular motion and connected objects p. 9

$$a_r = v^2/r$$
$$\sum F_{\text{inward}} = mv^2/r$$

7. Work and kinetic energy p. 10

$$W = Fd\cos\theta$$
$$K = \tfrac{1}{2}mv^2$$
$$W_{\text{net}} = K_f - K_i$$

8. Energy conservation and springs p. 11

$$K_i + U_i + W_{\text{other}} = K_f + U_f$$
$$U_g = mgh$$
$$U_s = \tfrac{1}{2}kx^2$$

9. Temperature, calorimetry and phase changes p. 13

$$Q = mc\Delta T$$
$$Q = mL$$
$$T_K = T_C + 273.15$$

10. Heat transfer and thermal expansion p. 14

$$\Delta L = \alpha L_0 \Delta T$$
$$P_{\text{cond}} = kA\Delta T/L$$
$$P_{\text{rad}} = \epsilon\sigma A(T^4 - T_s^4)$$

Notation

Symbols are listed by section. The same letter may name different quantities in different sections; read the section's Fact.

1. Choose units and describe motion

Δx	final position minus initial position (m)
Δt	elapsed time (s)
$\bar{v}$	average velocity (m/s)

2. Constant acceleration and graphs

u	initial velocity (m/s)
v	final velocity (m/s)
a	constant acceleration ($\mathrm{m/s^2}$)
s	displacement (m)
t	elapsed time (s)

3. Resolve vectors before adding

A_x	horizontal vector component (same as vector)
A_y	vertical vector component (same as vector)
R	resultant magnitude (same as vector)
θ	angle from the stated axis (degrees or radians)

4. Projectiles: one time, two directions

u_x	initial horizontal velocity (m/s)
u_y	initial vertical velocity (m/s)
g	gravitational acceleration magnitude ($\mathrm{m/s^2}$)
t	flight time (s)

5. Newton laws and friction

m	object mass (kg)
N	normal contact force (N)
μ_k	coefficient of kinetic friction (dimensionless)
a	acceleration ($\mathrm{m/s^2}$)

6. Circular motion and connected objects

v	speed along the circle (m/s)
r	circle radius (m)
a_r	inward acceleration ($\mathrm{m/s^2}$)
T	string tension (N)

7. Work and kinetic energy

W	work transferred by a force (J)
F	force magnitude (N)
d	displacement magnitude (m)
K	kinetic energy (J)

8. Energy conservation and springs

U_g	gravitational potential energy (J)
U_s	elastic potential energy (J)
k	spring stiffness (N/m)
x	spring deformation from natural length (m)

9. Temperature, calorimetry and phase changes

Q	energy transferred as heat (J)
c	specific heat capacity (J/(kg K))
L	specific latent heat (J/kg)
ΔT	temperature change (K or °C difference)

10. Heat transfer and thermal expansion

P	heat-transfer rate (W)
k	thermal conductivity (W/(m K))
A	transfer surface area (m^2)
L	wall thickness (m)
α	linear expansion coefficient (K^{-1})

Sources

Motion, vectors, forces, work, energy, temperature and heat. New merged lectures are the main mechanics source; the older heat lecture supplies thermal topics.

The source audit inventories 159 non-programming files; 95 yielded substantial extracted text. Scanned diagrams and handwritten or image-only material were not fully transcribed. Check lecturer-specific derivations and conventions. The practice tests are original study material, not predictions of an examination.

Index

Topics and quantities, with the page where each is introduced. Quantities show their symbol; the Index of notation lists symbols in alphabetical order.

A

acceleration (a), 8
angle from the stated axis (θ), 5
average velocity ($\bar{v}$), 2

C

Choose units and describe motion, 2
circle radius (r), 9
Circular motion and connected objects, 9
coefficient of kinetic friction (μ_k), 8
constant acceleration (a), 3
Constant acceleration and graphs, 3

D

displacement (s), 3
displacement magnitude (d), 10

E

elapsed time (Δt), 2
elapsed time (t), 3
elastic potential energy (U_s), 12
Energy conservation and springs, 11
energy transferred as heat (Q), 13

F

final position minus initial position (Δx), 2
final velocity (v), 3
flight time (t), 6
force magnitude (F), 10

G

gravitational acceleration magnitude (g), 6
gravitational potential energy (U_g), 12

H

Heat transfer and thermal expansion, 14
heat-transfer rate (P), 14
horizontal vector component (A_x), 5

I

initial horizontal velocity (u_x), 6
initial velocity (u), 3
initial vertical velocity (u_y), 6
inward acceleration (a_r), 9

K

kinetic energy (K), 10

L

linear expansion coefficient (α), 14

N

Newton laws and friction, 7
normal contact force (N), 8

O

object mass (m), 8

P

Projectiles: one time, two directions, 6

R

Resolve vectors before adding, 4
resultant magnitude (R), 5

S

specific heat capacity (c), 13
specific latent heat (L), 13
speed along the circle (v), 9
spring deformation from natural length (x), 12
spring stiffness (k), 12
string tension (T), 9

T

Temperature, calorimetry and phase changes, 13
temperature change (ΔT), 13
thermal conductivity (k), 14
transfer surface area (A), 14

V

vertical vector component (A_y), 5

W

wall thickness (L), 14
Work and kinetic energy, 10
work transferred by a force (W), 10

Index of notation

Latin letters first, then Greek letters, then differences (Δ). The same letter can name different quantities in different sections.

a acceleration, 8
a constant acceleration, 3
a_r inward acceleration, 9
A transfer surface area, 14
A_x horizontal vector component, 5
A_y vertical vector component, 5
c specific heat capacity, 13
d displacement magnitude, 10
F force magnitude, 10
g gravitational acceleration magnitude, 6
k spring stiffness, 12
k thermal conductivity, 14
K kinetic energy, 10
L specific latent heat, 13
L wall thickness, 14
m object mass, 8
N normal contact force, 8
P heat-transfer rate, 14
Q energy transferred as heat, 13
r circle radius, 9
R resultant magnitude, 5
s displacement, 3
t elapsed time, 3
t flight time, 6
T string tension, 9
u initial velocity, 3
u_x initial horizontal velocity, 6
u_y initial vertical velocity, 6
U_g gravitational potential energy, 12
U_s elastic potential energy, 12
$\bar{v}$ average velocity, 2
v final velocity, 3
v speed along the circle, 9
W work transferred by a force, 10
x spring deformation from natural length, 12
α linear expansion coefficient, 14
θ angle from the stated axis, 5
μ_k coefficient of kinetic friction, 8
Δt elapsed time, 2
ΔT temperature change, 13
Δx final position minus initial position, 2